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Fifty Years of Retracted Medical Publications From 1975 to 2024: A Comprehensive Analysis of Trends, Reasons, and Countries Using the Retraction Watch Database

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ABSTRACT

Background: Scientific medical research has progressed tremendously during the last 50 years, but concerns about research integrity, publishing ethics, and retraction trends have grown. Retractions are essential for revising the scientific record and maintaining credibility, yet an extensive long-term assessment of retracted medical publications is limited.

Methods: We performed a descriptive analysis of 50 years of retracted medical publications from the Retraction Watch Database. Data were refined to encompass solely medicine-related retractions, omitting corrections, expressions of concern, and reinstatements. We classified retraction reasons into 68 categories, emphasizing the top 10 most frequently encountered reasons. Temporal trends were evaluated employing semi-logarithmic linear regression models. The geographical distribution and journal-specific retractions were also examined.

Results: An analysis was conducted on 16,041 retracted medical documents from 1975 to 2024. The leading reasons for retraction included data concerns (31.47%), fraud (11.37%), peer review issues (11.21%), referencing issues (7.54%), and ethical violations (7.09%). The highest number of retractions was noted in *Computational and Mathematical Methods in Medicine* (5.91%), *Journal of Healthcare Engineering* (5.85%), and *Evidence-Based Complementary and Alternative Medicine* (4.36%). Approximately 45.28% of retracted papers included at least one author from China, followed by the United States and India. The medical subfields most impacted were oncology (19.87%), cardiovascular medicine (15.62%), and pharmacology (14.49%). Temporal analysis indicated a steady rise in retractions, with data concerns and fraud doubling typically every 5.5 and 5.2 years.

Conclusion: The rising amount of retractions underscores heightened scrutiny and enhanced detection techniques while highlighting ongoing research integrity issues. Data integrity, fraudulent activities, and compromised peer review are significant issues. Fortifying editorial

Data Sharing Statement

Raw data can be provided to researchers upon request.

Author Contributions

Conceptualization: Okyay RA, Kumar AB, Yessirkepov M, Sumbul HE, Kocyigit BF. Investigation: Okyay RA, Kocyigit BF. Methodology: Okyay RA, Kumar AB, Yessirkepov M, Sumbul HE, Kocyigit BF. Supervision: Kumar AB, Yessirkepov M, Sumbul HE. Writing - original draft: Okyay RA, Kumar AB, Yessirkepov M, Sumbul HE, Kocyigit BF. Writing - review & editing: Okyay RA, Kumar AB, Yessirkepov M, Sumbul HE, Kocyigit BF.

policies, augmenting transparency, and bolstering research ethics education are essential for reducing misconduct and maintaining the integrity of medical papers.

Keywords: Editorial Policies; Ethics; Medicine; Peer Review; Publishing; Research Ethics; Retraction Notice; Retraction of Publication

INTRODUCTION

Over the past fifty years, scientific medical research has achieved substantial advancements, resulting in transformative improvements in disease prevention, diagnosis, and treatment. The swift progress of medical knowledge has been propelled by innovations in biological technology, computational medicine, and evidence-based clinical practice.^{1,2} The enhancement in worldwide collaborations and the growth of open-access publishing have expedited the distribution of research results, making medical information more accessible.^{3,4} Nonetheless, although this advancement has enhanced medical outcomes and scientific discoveries, the increased published research also raised issues about the quality of studies, reliability, and ethical issues. The drive to publish, competition for financial support, and the changing dynamics of digital publishing have fostered a milieu in which scientific integrity may be undermined, either inadvertently through methodological inaccuracies or intentionally through unethical behavior.⁵⁻⁷ The increasing volume of medical publications has rendered the assurance of research findings' legitimacy and reliability an essential priority for the scientific world.

The retraction of scientific papers is essential to correct the medical literature and maintain research integrity. Retractions can arise due to numerous factors involving errors, ethical breaches, data falsification, plagiarism, and challenges associated with peer review.^{8,9} Consequences of misleading information reach significantly beyond individual papers, affecting scientific credibility, public confidence, and governmental decisions predicated on incorrect data. In medicine, where research directly impacts medical care and public health decisions, incorrect or misleading papers can have major consequences, potentially resulting in inefficient procedures, wasted funds, and patient harm.^{10,11} Comprehending trends in retractions is crucial for pinpointing gaps in the research environment, tackling misbehavior, and enhancing editorial and peer-review standards. Notwithstanding the increasing awareness of retractions, an extensive analysis of retracted medical articles over an extended time frame is still limited.¹²

This study seeks to perform a thorough temporal analysis of retracted medical papers over the last 50 years, identifying trends and patterns in retraction numbers. Through the comprehensive categorizing of retraction reasons, we aim to identify the predominant causes of retractions and evaluate their trend over various periods. Furthermore, we intend to examine retractions at the country and journal levels, offering insights into geographical and publication-related distribution. We attempt to identify the most affected disciplines by examining the distribution of retractions across various subfields of medical science. This comprehensive method seeks to advance the comprehension of retraction dynamics in medical research, aiding in enhancing research integrity and publishing standards.

METHODS

Study design

This descriptive study analyzed retracted publications from the Retraction Watch Database, covering 50 years up to 2024.

Data collection

We downloaded the complete Retraction Watch Database (<https://retractiondatabase.org/RetractionSearch.aspx?>) on March 1, 2025, containing 61,645 retraction-related records. The dataset was processed as follows:

1. Initial filtering: Medicine and medicine-related publications were extracted using a Python script (**Supplementary Data 1**), yielding 18,687 records.
2. Retraction type filtering: Records classified as “correction” (n = 465), “expression of concern” (n = 1,275), and “reinstatement” (n = 102) were excluded, retaining only retractions.
3. Deduplication: After importing data to SPSS, duplicate publications were removed based on title (n = 19).
4. Temporal validation:
 - Publications with zero days between publication and retraction were excluded due to potential data integrity issues (n = 668).
 - Retractions from 2025 were removed (n = 117).
5. Final dataset: The resulting dataset contained 16,041 retracted publications spanning from 1975 to 2024 (**Fig. 1**).

Categorization of retraction reasons

We used another Python script to extract unique retraction reason categories (**Supplementary Data 2**). A total of 106 unique categories were formed out of 16,041 retracted publications. The Retraction Watch Database includes some information regarding the investigating party, the status of publication after retraction, and the nature of and quality of retraction notice among the reason categories (Retraction Watch Database User Guide Appendix B: Reasons available at: <https://retractionwatch.com/retraction-watch-database-user-guide/retraction-watch-database-user-guide-appendix-b-reasons/>), so we excluded them. Finally, 68 categories were presented. Some of the categories shared a common retraction reason. Thus, we have presented them together, without merging but summing up the counts. We also defined the top 10 main retraction reasons and used these to visualize trends over time:

- Data concerns: Issues with data integrity or unreliable results.
- Fraud: Deliberate deception in research conduct, including falsified experiments or manipulated data.
- Peer review issues: Compromised peer review process, including fake reviews or reviewer manipulation.
- Referencing issues: Improper citation practices or reference manipulation to increase impact.
- Ethical issues: Violations of research ethics, including consent problems.
- Duplication: Publication of substantially overlapping content in multiple journals.
- Error: Genuine mistakes in methodology, analysis, or interpretation requiring retraction.
- Plagiarism: Unattributed use of others’ work, ideas, or language presented as original.
- Randomly generated content: Articles created using artificial intelligence tools without proper review or verification.
- Misconduct: General research misconduct not specified in other categories.

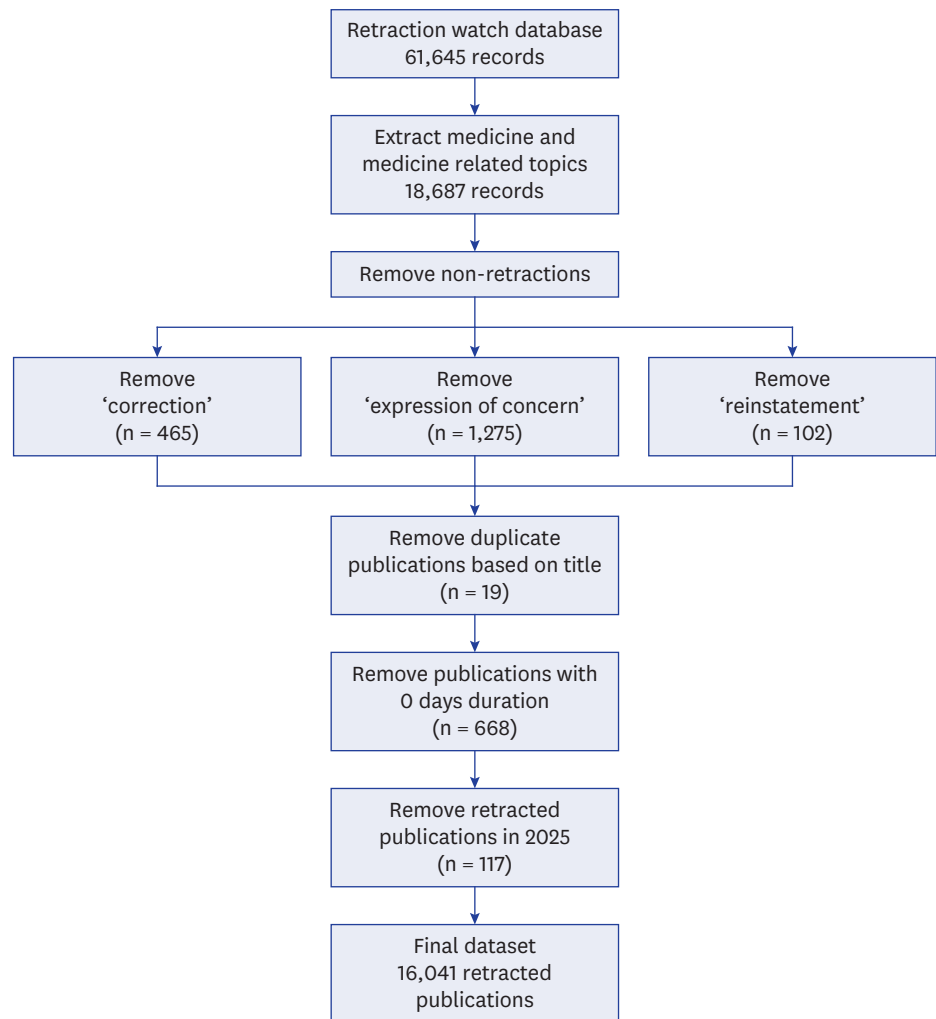


Fig. 1. Visualization of data acquisition from Retraction Watch Database.

Statistical analysis

Descriptive statistics were presented as numbers, percentages, means, standard deviations, medians, and minimum and maximum. To analyze trends in retractions over time, we performed linear regression analysis with year as the independent variable and number of retractions as the dependent variable. Due to the low model fit of the initial linear model, we applied a semi-logarithmic linear regression model using statsmodels library in Python with the natural logarithm of retracted publication counts, which provided a significantly better fit. A correlation matrix was constructed. Spearman's rank correlation analysis was employed to examine the relationships between retraction reasons and year variable. This non-parametric correlation method was chosen to assess monotonic relationships between variables. The year variable was included in the analysis to evaluate the temporal trends of retractions. The scipy.stats module in Python was used to calculate the correlation coefficients. Statistical significance was set at $P < 0.05$, and results were marked at three different significance levels ($P < 0.05$, $P < 0.01$, $P < 0.001$).

For geographical analysis, we created a choropleth map using Natural Earth, a public domain map dataset, geopandas and matplotlib libraries in Python (**Supplementary Data 3**) to visualize the global distribution of retracted publications by country.

PubMed indexing rates were determined by checking the availability of either a retraction notice or publication PubMed ID (PMID) in the database. Web of Science (WoS) indexing rates were assessed by converting existing article Digital Object Identifiers (DOIs) into batched queries and systematically searching the Web of Knowledge site. A χ^2 analysis was performed to examine the distribution of retracted articles between WoS-indexed and non-WoS journals using the `scipy.stats` module in Python.

Ethics statement

This study utilized publicly available data from the Retraction Watch Database without involving human subjects, patient data, or experimental procedures, so formal ethical approval was not required.

RESULTS

According to the retraction database, the first retracted publication date was 3/7/1975 and last retracted publication date was 30/12/2024. The median time for the articles to remain in publication before retraction was 562 (1–29,622) days. There were 48,649 retraction reason counts and 106 unique categories in total. The most common retraction reasons were recorded as data concerns (31.47%), fraud (11.37%), peer review issues (11.21%), referencing issues (7.54%), and ethical issues (7.09%), respectively (**Table 1**).

The journals that retracted the most publications were *Computational and Mathematical Methods in Medicine* (5.91%), *Journal of Healthcare Engineering* (5.85%), and *Evidence-Based Complementary and Alternative Medicine* (4.36%). The database had 3,175 unique sources, and 29.87% of the total retracted publications (16,041) were retracted by the 10 sources presented in **Table 2**.

The countries from which the most retractions originated were summarized in **Table 3**. Most retracted publications were from China, followed by the United States and India. Strikingly, nearly half of the retracted publications (45.28%) were authored by at least one researcher from China. A comprehensive heatmap showing the geographical distribution of retracted

Table 1. Distribution of retraction reasons and subcategories

Retraction category	Retraction subcategory	No. (%) ^a	Total, No. (%) ^b
Data concerns	Unreliable results	6,016 (39.29)	15,311 (31.47)
	Concerns/issues about data	4,895 (31.97)	
	Concerns/issues about results	2,756 (18.00)	
	Concerns/issues about image	662 (4.32)	
	Unreliable data	579 (3.78)	
	Results not reproducible	264 (1.72)	
	Unreliable image	74 (0.48)	
	Contamination of cell lines/tissues	44 (0.29)	
	Contamination of materials (general)	21 (0.14)	
Fraud	Paper mill	3,476 (63.08)	5,531 (11.37)
	Falsification/fabrication of data	755 (13.65)	
	False/forged authorship	380 (6.87)	
	Manipulation of images	366 (6.62)	
	Rogue editor	255 (4.61)	
	Falsification/fabrication of image	180 (3.25)	
	Falsification/fabrication of results	60 (1.08)	
	False affiliation	32 (0.58)	
	Manipulation of results	22 (0.40)	
	Hoax paper	5 (0.09)	

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Table 1. (Continued) Distribution of retraction reasons and subcategories

Retraction category	Retraction subcategory	No. (%) ^a	Total, No. (%) ^b
Peer review issues	Concerns/issues about peer review	3,489 (63.98)	5,452 (11.21)
	Fake peer review	1,963 (36.02)	
Referencing issues	Concerns/issues about referencing/attribution	3,582 (97.60)	3,670 (7.54)
	Cites retracted work	88 (2.40)	
Ethical issues	Lack of IRB/IACUC approval	1,643 (47.61)	3,451 (7.09)
	Informed/patient consent - none/withdrawn	949 (27.50)	
	Concerns/issues about human subject welfare	397 (11.50)	
	Ethical violations by author	335 (9.71)	
	Lack of approval from the company/institution	97 (2.81)	
	Ethical violations by third party	23 (0.67)	
	Concerns/issues about animal welfare	7 (0.20)	
Duplication	Duplication of article	1,010 (32.89)	3,071 (6.31)
	Duplication of image	937 (30.51)	
	Euphemisms for duplication	379 (12.34)	
	Duplicate publication through error by journal/publisher	371 (12.08)	
	Duplication of data	213 (6.94)	
Error	Duplication of text	161 (5.24)	2,801 (5.76)
	Error in data	717 (25.60)	
	Error in analyses	466 (16.64)	
	Error in methods	460 (16.42)	
	Error in results and/or conclusions	428 (15.28)	
	Error in image	272 (9.71)	
	An error by journal/publisher	185 (6.60)	
	Error in text	157 (5.60)	
	Error in materials	59 (2.11)	
	Error in cell lines/tissues	33 (1.18)	
Plagiarism	Error by third party	24 (0.86)	2,245 (4.61)
	Plagiarism of article	716 (31.89)	
	Euphemisms for plagiarism	680 (30.29)	
	Plagiarism of text	480 (21.38)	
	Plagiarism of image	187 (8.33)	
	Plagiarism of data	131 (5.84)	
	Taken from dissertation/thesis	46 (2.05)	
Randomly generated content	Taken via peer review	5 (0.22)	2,161 (4.44)
Misconduct	Misconduct by author	1,000 (52.97)	1,888 (3.88)
	Misconduct - official investigation/finding	862 (45.66)	
	Misconduct by third party	16 (0.85)	
	Euphemisms for misconduct	7 (0.37)	
	Misconduct by company/institution	3 (0.16)	
Transparency issues	Original data not provided	541 (47.41)	1,141 (2.35)
	Author unresponsive	600 (52.59)	
Authorship issues	Concerns/issues about authorship/affiliation	487 (70.07)	695 (1.43)
	Lack of approval from the author	208 (29.93)	
Third-party issues	Concerns/issues about third-party involvement	157 (51.31)	306 (0.63)
	Lack of approval from third party	149 (48.69)	
Breach of policy by author			209 (0.43)
Copyright claims			181 (0.37)
Conflict of interest			158 (0.32)
Others			378 (0.78)
Total^c			48,649 (100)

IRB = Institutional Review Board, IACUC = Institutional Animal Care and Use Committee.

^aRow percentage; ^bColumn percentage; ^cMultiple reasons were available for individual articles.

publication intensity is presented in **Fig. 2**. Top medicine topics that are related to retraction were oncology (19.87%), followed by cardiovascular medicine (15.62%) and pharmacology (14.49%). Detailed distribution is presented in **Table 4**.

Table 2. Top 10 sources of retracted publications

Source of retracted publications	No. (%)
<i>Computational and Mathematical Methods in Medicine</i>	948 (5.91)
<i>Journal of Healthcare Engineering</i>	938 (5.85)
<i>Evidence-Based Complementary and Alternative Medicine</i>	699 (4.36)
<i>Cochrane Database of Systematic Reviews</i>	571 (3.56)
<i>BioMed Research International</i>	554 (3.45)
<i>European Review for Medical and Pharmacological Sciences</i>	241 (1.50)
<i>Computational Intelligence and Neuroscience</i>	224 (1.40)
<i>Contrast Media & Molecular Imaging</i>	222 (1.38)
<i>PLoS One</i>	217 (1.35)
<i>Applied Bionics and Biomechanics</i>	177 (1.10)

Table 3. Top 10 countries of retracted publications

Country of retracted publications	Values ^a
China	7,264 (45.28)
United States	2,268 (14.13)
India	1,096 (6.83)
Japan	896 (5.58)
United Kingdom	779 (4.85)
Germany	604 (3.76)
Saudi Arabia	491 (3.06)
Iran	391 (2.43)
South Korea	379 (2.36)
Italy	378 (2.35)

Values are presented as number (%).

^aPublications authored by multiple countries were also counted. Thus, such articles were included in more than one entry.

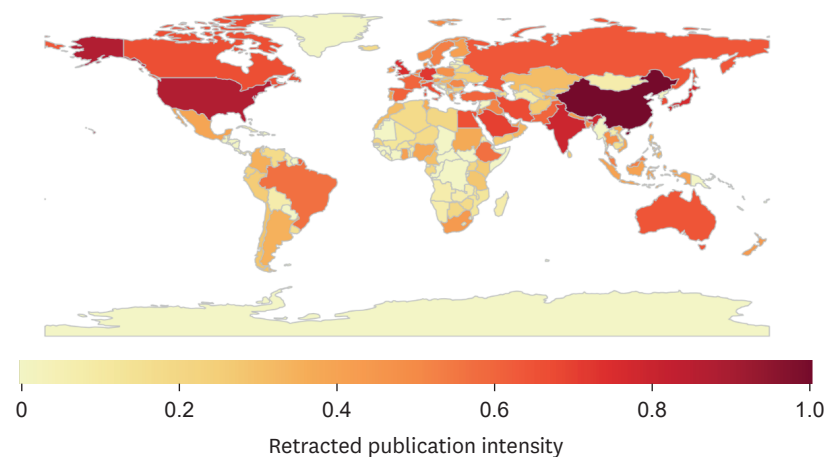


Fig. 2. Geographical distribution of retracted publication intensity.

In the Retraction Watch Database, multiple article types may be attributed to a single publication. The database contained 16,908 entries for the 16,041 publications included in this study. Research articles constituted the majority of retracted publications (62.21% of entries), followed by clinical studies (14.08%) and reviews (7.99%).

Of the 16,041 publications examined, 13,776 (85.87%) had either a retraction or publication PMID, and 15,779 (98.36%) had DOI numbers. We structured WoS query batches using these DOI numbers and discovered that 12,568 (78.35%) of the retracted publications were indexed in WoS. A χ^2 analysis assessed the distribution of retracted articles in WoS-indexed and non-

Table 4. Top 10 medicine topics of retracted publications

Topics	Values ^a
Oncology	3,187 (19.87)
Cardiovascular ^b	2,506 (15.62)
Pharmacology	2,324 (14.49)
Surgery	1,983 (12.36)
Neurology	1,864 (11.62)
Obstetrics/gynecology	1,462 (9.11)
Rehabilitation/therapy	1,380 (8.60)
Orthopedics	1,244 (7.76)
Gastroenterology	1,216 (7.58)
Infectious disease	1,141 (7.11)

Values are presented as number (%).

^aMany publications were attributed to multiple topics; thus, such articles were included in more than one entry.

^bCardiology and cardiovascular topics were merged.

WoS sources. The data showed that 12,575 articles were published in 2,230 WoS sources, while 3,466 articles appeared in 945 non-WoS journals. The chi-square test revealed a significant difference in this distribution ($\chi^2 = 99.7048$; $df = 1$; $P < 0.001$). The odds ratio of 1.5375 indicates that WoS-indexed journals have approximately 1.54 times higher proportion of retracted articles than non-WoS journals. On average, WoS journals contained 5.6390 retracted articles per journal, while non-WoS journals averaged 3.6677 retracted articles per journal.

Using line graphs, we visualized the retraction counts for the most common reasons for retraction. All main reasons for retraction showed an upward trend. We employed a semi-logarithmic linear regression analysis using the natural logarithm of retraction counts of the most common three retraction reasons over time. The annual growth rate for data concerns was estimated at 12.60% ($B = 0.126$; 95% confidence interval [CI], 0.107–0.146; $P < 0.001$; $R^2 = 0.802$), with retractions approximately doubling every 5.50 years. For fraud, the growth rate was slightly higher at 13.30% ($B = 0.133$; 95% CI, 0.113–0.152; $P < 0.001$; $R^2 = 0.838$), with retractions doubling every 5.21 years. However, while we found an annual rate of 29.70% for peer review issues, there was no statistical significance (Fig. 3).

In addition, a correlation matrix was constructed. The matrix revealed significant positive correlations ($P < 0.05$) between most retraction reasons, suggesting these issues frequently co-occur. Particularly strong correlations ($r > 0.9$) existed between years, duplication, data concerns, and fraud. Peer review issues demonstrated fewer significant correlations with other retraction reasons (Fig. 4).

DISCUSSION

This study offers an extensive temporal analysis of retracted medical papers over the last 50 years, emphasizing patterns in retractions, reasons, geographical distributions, and medical subfields. Our analyses show that data concerns, fraud, and peer review challenges are the most prominent causes. The study also highlights disparities in retraction numbers across countries and journals, with a clear concentration of retractions originating from specific countries and sources. Oncology, cardiovascular medicine, and pharmacology emerged as the most commonly impacted medical subfields.

Our analysis revealed that data issues were the predominant reason for retraction, comprising 31.47% of instances. This emphasizes the critical importance of data integrity in

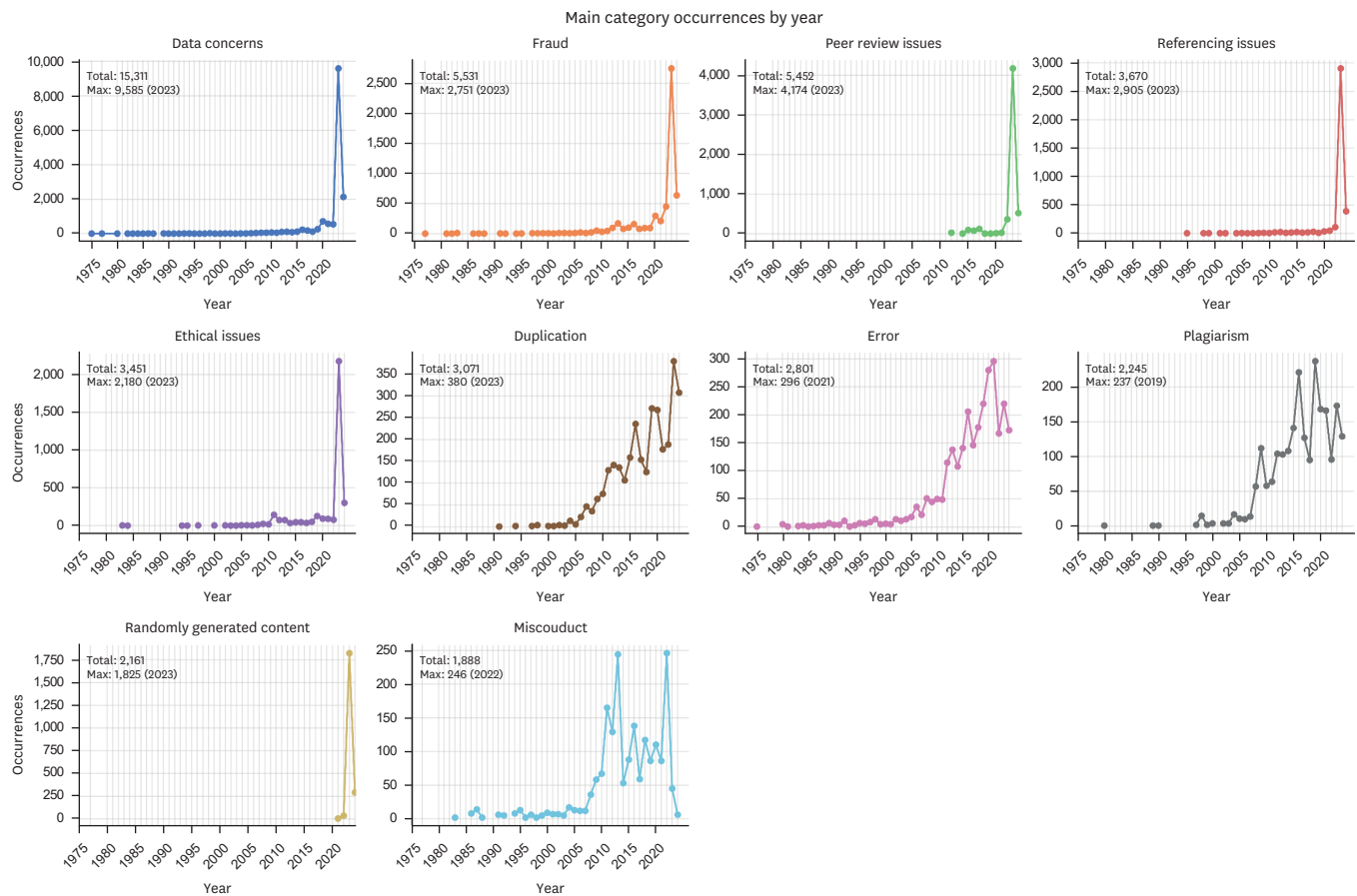


Fig. 3. Temporal analysis of retraction reason.

Variables	Year	Data concerns	Duplication	Error	Ethical issues	Fraud	Misconduct	Peer review issues	Plagiarism	Randomly generated content	Referencing issues
Year	1.000	0.904***	0.965***	0.948***	0.902***	0.945***	0.730***	0.490	0.907***	0.800	0.941***
Data concerns	0.904***	1.000	0.973***	0.852***	0.898***	0.922***	0.705***	0.529	0.920***	0.800	0.916***
Duplication	0.965***	0.973***	1.000	0.954***	0.901***	0.956***	0.684***	0.445	0.925***	1.000***	0.871***
Error	0.948***	0.852***	0.954***	1.000	0.876***	0.924***	0.742***	-0.053	0.933***	-0.200	0.874***
Ethical issues	0.902***	0.898***	0.901***	0.876***	1.000	0.907***	0.661***	0.273	0.789***	0.800	0.878***
Fraud	0.945***	0.922***	0.956***	0.924***	0.907***	1.000	0.717***	0.677*	0.864***	1.000***	0.929***
Misconduct	0.730***	0.705***	0.684***	0.742***	0.661***	0.717***	1.000	-0.191	0.635***	-0.600	0.623***
Peer review issues	0.490	0.529	0.445	-0.053	0.273	0.677*	-0.191	1.000	-0.025	1.000***	0.644*
Plagiarism	0.907***	0.920***	0.925***	0.933***	0.789***	0.864***	0.635***	-0.025	1.000	0.400	0.748***
Randomly generated content	0.800	0.800	1.000***	-0.200	0.800	1.000***	-0.600	1.000***	0.400	1.000	1.000***
Referencing issues	0.941***	0.916***	0.871***	0.874***	0.878***	0.929***	0.623***	0.644*	0.748***	1.000***	1.000

Significance: * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$

Color coding: ■ $R > 0.7^*$ ■ $0.5 < R \leq 0.7^*$ ■ $0.3 < R \leq 0.5^*$ ■ $P \geq 0.05$

Fig. 4. Correlation matrix including retraction reasons and year.

medical research. Data mining, statistical assessment, or repeatability issues can undermine study validity, resulting in retractions. The heightened focus on data transparency and reproducibility may have facilitated the rising identification of such vulnerabilities.^{13,14}

Fraud was the second most common reason for retractions. The competitive landscape of academic publishing and the pressure to provide high-impact research may compel certain researchers to engage in unethical practices. The increasing identification of fraud may also indicate heightened scrutiny of journal editors, peer reviewers, and independent oversight groups.¹⁵

Peer review issues comprised 11.21% of retractions, highlighting tricks like fake reviews or corrupted review systems. Such incidents reveal flaws in the peer review system, particularly in journals with insufficient editorial control. As predatory journals expand, peer review integrity continues a major challenge.¹⁶

Referencing issues (7.54%) reflect that inappropriate citation processes, citation inflation, and reference engineering continue to damage the trustworthiness of research. Such factors can bias the perceived impact of research and jeopardize academic integrity.¹⁷

Ethical issues (7.09%) raise concerns about scientific misconduct, including failure to obtain ethics committee approval, absence of informed consent, conflicts of interest, and inappropriate research methodology. Ethical violations in medicine are of special concern since they can directly impact patient safety and public trust in scientific findings.¹⁸

Our data showed that a substantial percentage of retractions stemmed from a limited number of journals. *Computational and Mathematical Methods in Medicine* (5.91%), *Journal of Healthcare Engineering* (5.85%), and *Evidence-Based Complementary and Alternative Medicine* (4.36%) had the highest number of retractions. Furthermore, just 10 journals constituted around 30% of all retractions, underscoring the uneven distribution of retracted publications among sources. This proves that retractions are not evenly distributed throughout medical journals but are concentrated in particular settings, potentially owing to differences in editorial practices, peer review stringency, or historical susceptibilities to compromised research integrity.¹⁹

The analysis identified China as the leading source of retracted papers, followed by the United States and India. Notably, 45.28% of all retracted papers included at least one author from China, signifying a deep concentration of retractions in this area. The dominant ranking of China in retracted products might be ascribed to several variables. In recent years, China has become a global leader in scientific research output, characterized by an exponential rise in medical publications. The substantial number of submissions inherently elevates the probability of retractions. The Chinese academic structure prioritizes the quantity of papers for career progression, funding acquisition, and institutional placements.²⁰ The rigorous “publish or perish” mentality may result in heightened occurrences of scientific misconduct, such as data falsification and plagiarism.²¹

In addition to individual misconduct and editorial shortcomings, socio-political and institutional factors may influence retractions, particularly in research-intensive countries. Systemic academic incentives, such as performance-based promotions, publication quotas linked to graduation or job placements, and financial rewards for publishing in indexed journals, might unintentionally encourage unethical research. The rapid growth of academic

institutions and inadequate regulatory frameworks can compromise peer review, editorial standards, and training in research ethics within swiftly changing research environments, especially in emerging or transitional nations. Political influences, encompassing governmental directives to enhance international visibility through publications, may prioritize quantity over quality. As our findings suggest, these systemic issues may elucidate the observed clustering of retractions in specific journals and regions.^{22,23}

The analysis of retracted papers among medical subfields indicated that oncology had the highest number of retractions, followed by cardiovascular medicine and pharmacology. This pattern emphasizes particular fields of medical research where retractions are more frequent and prompts inquiries into the root causes. Cancer research attracts substantial international funding and interest, resulting in a considerable number of publications.²⁴ The extensive number of studies may enhance the probability of identifying errors or misconduct. These disciplines commonly involve complicated methodologies, encompassing biomarker discovery, clinical trials, genetic assessments, extensive clinical cohorts, and long-term investigations. Thus, defects in study design, statistical analysis, or reproducibility may lead to retractions.²⁵

Over time, the analysis of retraction patterns demonstrated an upward trend for all main reasons for retraction. This trend highlights the escalating identification and documentation of misleading publications in medical research. Results regarding data concerns suggest a growing identification of issues relating to data integrity, unreliable outcomes, and incorrect methodologies. Factors contributing to increased fraud detection may encompass intensified scrutiny of journals and advanced digital tools for identifying manipulation.^{26,27}

The consistent rise in retractions across all categories underscores the heightened vigilance of the scientific community in upholding research integrity. The enhancement in identifying fraudulent and unreliable research underscores the necessity for more robust preventive strategies, including intensified peer review procedures, transparent data dissemination, and more powerful institutional supervision. The swift increase in retractions related to peer review indicates a necessity for enhanced editorial policies and verification systems to avert fraudulent or compromised review processes. The rising number of retractions reflects both the growing scrutiny of medical research and the persistent difficulty of preserving publication integrity.²⁸ Resolving these difficulties necessitates proactive initiatives in research education, journal regulations, and improvements in peer review to avert misbehavior.²⁹

The article has several limitations. The research depends on data from the Retraction Watch Database, which, although extensive, may not encompass all retracted papers, particularly those from weak-indexed or local journals. The data solely reflects medicine-related retractions. Retraction trends may be affected by adjustments in detection methodologies, journal regulations, or reporting processes rather than indicating genuine fluctuations in research misconduct frequencies.

The strengths are listed as follows. The large sample size offers considerable statistical power and facilitates rigorous examination of trends across several medical subfields. The systematic filtering process, characterized by explicit inclusion and exclusion criteria, improves the reproducibility of the study methods.

This study offers a comprehensive review of retracted medical papers from the previous 50 years, highlighting notable trends in the reasons for retraction, geographical distribution,

and impacted medical subfields. The rising number of retractions, especially attributed to data concerns, fraud, and peer review deficits, highlights the necessity for ongoing enhancements in research integrity, editorial supervision, and peer review processes. The increasing frequency of retractions indicates enhanced scrutiny and improved detection techniques, although it also reveals ongoing vulnerabilities in scientific publishing.

Journals, academic organizations, and institutions can implement numerous practical measures to mitigate fraudulent publications and foster a culture of research integrity. Journals should adopt more stringent editorial practices by integrating comprehensive screening procedures to detect plagiarism, image manipulation, and fabricated data and utilizing advanced software tools before the peer review process. Editors may benefit from receiving formal training to recognize indicators of scientific misconduct and employing standardized checklists throughout the article evaluation process. Enhancing the peer-review process is essential; utilizing verified reviewers' databases, implementing double-blind or open peer-review procedures, and avoiding reliance on reviewers suggested by authors can reduce manipulation. Furthermore, academic institutions and universities should provide comprehensive training in research ethics to all graduate and postgraduate students, emphasizing case-based learning, transparency, data management, and publishing ethics.

Fostering a shift in academic culture—from metrics focusing on output to assessments prioritizing quality and integrity—can significantly reduce the pressures that contribute to unethical behavior. Employing multi-layered strategies is essential for maintaining the long-term reliability of medical research.

SUPPLEMENTARY MATERIALS

Supplementary Data 1

Python script for filtering and analyzing medicine-related publications from the Retraction Watch Database

Supplementary Data 2

Python script for categorizing and processing retraction reasons

Supplementary Data 3

Python script for geographical analysis and heatmap visualization of retracted publications

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